$$\frac{\Delta \mid \Gamma \,;\, \Omega_A \vdash e_1 : A \quad \Delta \mid \Gamma \,;\, \Omega_B \vdash e_2 : B}{\Delta \mid \Gamma \,;\, \Omega_B \, \Omega_A \vdash (e_1, e_2) : A \circ B} \circ I$$

$$\frac{\Delta \mid \Gamma \,;\, \Omega \vdash e : A \circ B \quad \Delta \mid \Gamma \,;\, \Omega_L \, (y : B) \, (x : A) \, \Omega_R \vdash e' : C}{\Delta \mid \Gamma \,;\, \Omega_L \, \Omega \, \Omega_R \vdash \mathbf{match}\ e\ ((x, y) \Rightarrow e') : C} \circ E$$

$$\frac{(k \in L) \quad \Delta \mid \Gamma \,;\, \Omega \vdash e : A_k}{\Delta \mid \Gamma \,;\, \Omega \vdash k(e) : \oplus\{\ell : A_\ell\}_{\ell \in L}} \oplus I$$

$$\frac{\Delta \mid \Gamma \,;\, \Omega \vdash e : \oplus\{\ell : A_\ell\}_{\ell \in L} \quad (\Delta \mid \Gamma \,;\, \Omega_L \, (x_\ell : A_\ell) \, \Omega_R \vdash e_\ell : A_\ell) \quad (\forall \ell \in L)}{\Delta \mid \Gamma \,;\, \Omega_L \, \Omega \, \Omega_R \vdash \mathbf{match}\ e\ \{\ell(x_\ell) \Rightarrow e_\ell\}_{\ell \in L} : C} \oplus E$$